Landmark Measurement Model

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1. Overview

> **Key point:** A landmark reading is a range and a bearing to a known point, and the model scores a guessed pose with two **Gaussians** (G-827) around the range and bearing that pose predicts. One reading puts the robot on a ring, three readings pin it to one spot, and a reading can also be turned around to propose poses.

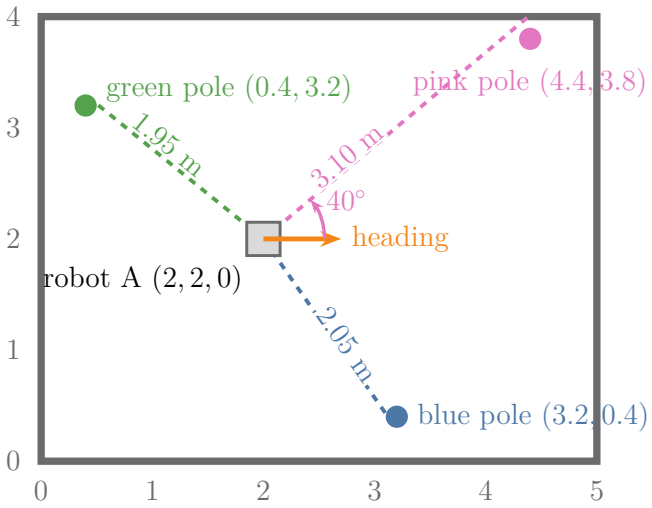


Figure 1: The 5 m by 4 m room with three coloured poles: pink at (4.4, 3.8), green at (0.4, 3.2), blue at (3.2, 0.4). The robot at A = (2, 2, 0) reads each one as a range and a bearing: pink 3.10 m at 40 degrees, green 1.95 m at 141 degrees, blue 2.05 m at −55 degrees

The **likelihood field** (G-2393) lets the robot score a laser scan about 250 times faster than the beam model. What still breaks: a scan gives hundreds of distances, all of which look alike: wall, wall, wall. This Note adds readings that say which object was seen. A camera that spots a pink pole gives far less data, but the data says exactly which object it saw, and the map says exactly where that object stands. Such a recognisable object at a known place is a landmark. Robots pull landmarks out of sensor data, and each one is seen as a range (how far), a bearing (at what angle from the robot's **heading** (G-2284)) and a signature (what it looks like, here its colour).

Figure 1 shows the test case of this Note: the same 5 m by 4 m room and pose A = (2, 2, 0) as the **beam model** (G-2377), now with three coloured poles that the robot sees. To use these readings, the robot needs the same thing the **beam model** (G-2377) gave it for a laser: a measurement model (G-2374), the density of the readings $p(z \mid x, m)$ for any guessed pose x in the map m. The question is: how well does a guessed pose, such as A or a pose B 1 m to its right, explain these three readings? This Note builds the model that answers it:

- what range and bearing a pole should show from a guessed pose, and the two pitfalls in computing the angle (Section 3);

- the density of a reading, and why it is a product of two Gaussians (Section 4);
- why one reading puts the robot on a ring, and several readings pin it down (Section 5);
- how to turn one reading around and draw poses that fit it (Section 6).

2. A landmark reading

> **Key point:** Each reading is a range, a bearing and a signature; the signature tells us which landmark of the map was seen.

First, what the robot at A gets from each pole of Figure 1. We write one landmark reading as a list of three entries, the range, bearing and signature (G-2343, G-2344, G-2345):

- r is the range: the distance from the robot to the landmark, in m (3.10 m for the pink pole);
- φ (phi) is the bearing: the angle from the robot's heading to the landmark, counter-clockwise positive, as for the heading (G-2284) itself, in degrees (40° for the pink pole);
- s is the signature: what identifies the landmark, here its colour (pink);
- z is the whole reading, the three entries together.

$$z = (r,\ \varphi,\ s)$$

The three readings of Figure 1:

Pole	Range r	Bearing φ	Signature s
pink	3.10 m	40°	pink
green	1.95 m	141°	green
blue	2.05 m	−55°	blue

A sensor that gives both numbers, such as a laser that picks poles out of its scan or a calibrated camera that also judges size, is a range-bearing sensor (Brown CS148 tutorial; Freiburg sensor-model slides, "Landmarks"). The signature solves a problem the laser scan never had to face: which landmark of the map is this? Knowing which map landmark a reading belongs to is **known correspondence** (G-2405). Here each colour appears once, so the colour gives the correspondence directly (Brown CS148 tutorial, §3). For the landmark a reading belongs to:

- c is its index, the number of that landmark in the map's list;
- $m_{c,x}$ is its x coordinate in the map, in m (4.4 m for the pink pole);
- $m_{c,y}$ is its y coordinate in the map, in m (3.8 m for the pink pole).

For the pink pole:

$$(m_{c,x},\ m_{c,y}) = (4.4,\ 3.8)$$

When colours repeat, the correspondence is unknown and must be worked out too. Here each pole is known, so for any guessed pose the robot can ask what that pole should read.

3. What a guessed pose predicts

> **Key point:** From a guessed pose, position (x, y) and **heading** (G-2284) θ, a landmark should appear at the straight-line distance and at its direction minus the heading. Compute the direction with atan2, and wrap every angle difference into $-180°$ to $180°$.

3.1 Expected range and bearing

To score a guessed pose against the three readings of Figure 1, we first ask what the robot would read from there. The steps are those that turn a known pose and the map into a reading, now done from a guessed pose. Figure 2 draws the pink pole seen from (2, 2), with the robot turned 15° so that the direction to the pole (green) and the bearing (pink) can be told apart; at pose A the heading is 0° and the two are equal. The pole lies 2.4 m further along x and 1.8 m further along y:

- Δx is how far the pole lies beyond the robot along x, in m;
- Δy is how far it lies beyond the robot along y, in m.

$$\Delta x = 4.4 - 2 = 2.4$$

$$\Delta y = 3.8 - 2 = 1.8$$

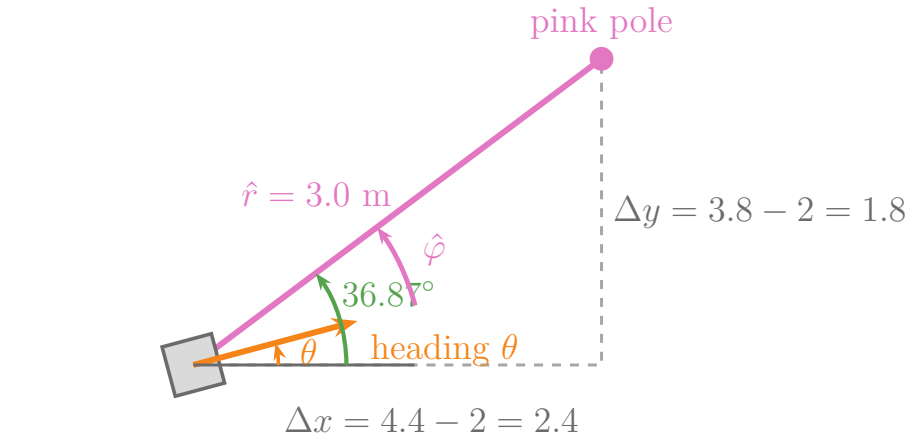


Figure 2: The pink pole seen from a robot at (2, 2): a right-angled triangle with sides 2.4 along x and 1.8 along y and long side 3.0 m. The direction to the pole is 36.87 degrees from the x-axis; the bearing is that direction minus the robot's heading

Expected range. The straight-line distance is the long side of the triangle (Pythagoras), the Euclidean distance (G-715). A hat marks a predicted value:

- $\hat{r}$ is the expected range, the distance the robot should read, in m.

$$\hat{r} = \sqrt{\Delta x^2 + \Delta y^2}$$

$$= \sqrt{5.76 + 3.24} = 3.0 \text{ m}$$

Expected bearing. The direction from the robot to the pole, measured from the world x-axis, is the angle whose tangent is $\Delta y / \Delta x$:

$$\text{direction} = \text{atan2}(1.8,\ 2.4) = 36.87°$$

The bearing is measured from the robot's **heading** (G-2284), not from the x-axis, so we subtract the heading (Figure 2):

- θ is the robot's heading, in degrees (0° at pose A);
- $\hat{\varphi}$ is the expected bearing, in degrees.

$$\hat{\varphi} = 36.87° - \theta$$

$$= 36.87^\circ - 0^\circ = 36.87^\circ$$

In general, for landmark c and any guessed pose:

- x is the guessed position's x coordinate, in m (2 m at A);
- y is its y coordinate, in m (2 m at A);
- $m_{c,x}$ is landmark c's x coordinate in the map, in m (4.4 m for the pink pole);
- $m_{c,y}$ is its y coordinate in the map, in m (3.8 m for the pink pole).

Together, the **expected range and bearing** (G-2402) of landmark c from pose (x, y, θ) are (Thrun §6.6; Brown CS148 tutorial, Eq. 5–6):

$$\hat{r} = \sqrt{(m_{c,x} - x)^2 + (m_{c,y} - y)^2}$$

$$\hat{\varphi} = \text{atan2}(m_{c,y} - y,\ m_{c,x} - x) - \theta$$

3.2 Why atan2 and not arctan

The expected bearing of Section 3.1 takes the direction from atan2, not from the plain arctangent of the ratio. The green pole of Figure 1, at (0.4, 3.2), shows why. From A, how far it lies beyond the robot along x, Δx, and along y, Δy:

$$\Delta x = 0.4 - 2 = -1.6$$

$$\Delta y = 3.2 - 2 = 1.2$$

The plain arctangent of the ratio:

$$\arctan(1.2/-1.6) = \arctan(-0.75)$$

$$= -36.87^\circ$$

That direction points down and to the right, away from the pole (Figure 3). The ratio has lost the signs: a point at (1.6, −1.2) gives the same ratio −0.75. So the formula uses atan2 (G-2325), which takes Δy and Δx separately, so it knows which quarter of the plane the point lies in, and returns the angle in the right quarter, between −180° and 180° (Brown CS148 tutorial, §3):

$$\text{atan2}(1.2,\ -1.6) = 143.13^\circ$$

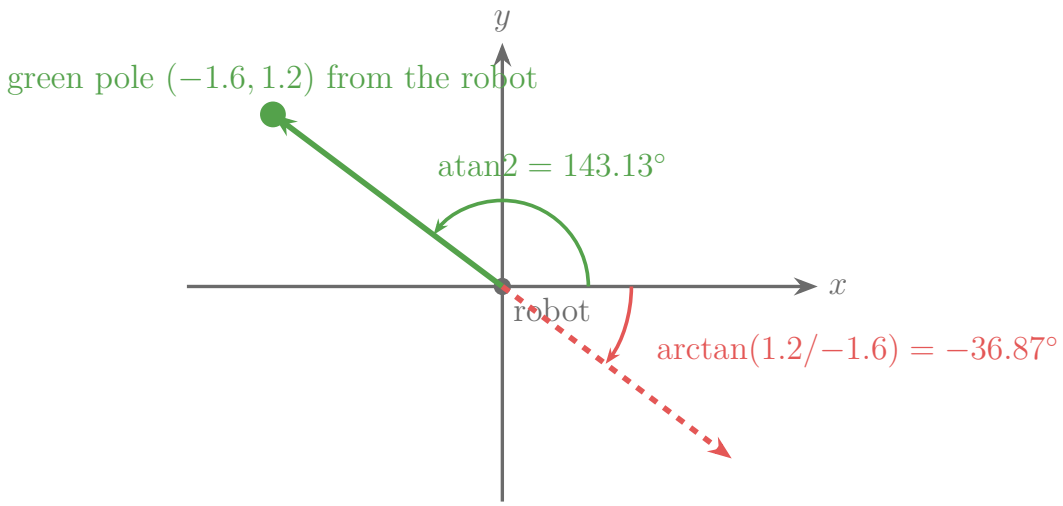


Figure 3: The green pole, at (−1.6, 1.2) as seen from the robot. atan2 gives 143.13 degrees, pointing up and left at the pole; arctan of the ratio 1.2 / −1.6 gives −36.87 degrees, pointing down and right, the opposite way

3.3 Why angle differences must be wrapped

Scoring a reading means comparing it with the prediction. For range, the difference is a plain subtraction. For bearings it is not. None of the three poles sits near the ±180° line, where the trouble shows, so take a made-up case: the bearing reads 175° and the prediction is −178°. The two directions are only 7° apart, across the ±180° line, but the plain difference is:

$$175 - (-178) = 353°$$

A bell-shaped error curve around the prediction, a **Gaussian** (G-827), would call that a wild miss. The fix is to add or subtract 360° until the difference lies between −180° and 180°:

$$353° - 360° = -7°$$

Adding or subtracting full turns in this way is **angle wrapping** (G-2404), the wrap-around of **headings** (G-2284) on a dial, where 350° and 10° are 20° apart. Every bearing difference and every heading in this Note is wrapped (Brown CS148 tutorial, §3, which uses the angle between two unit vectors to the same effect). With the prediction and a fair way to subtract angles, the robot can now score how far each reading lies from what pose A predicts.

4. The landmark measurement model

> **Key point:** The density of a reading is a **Gaussian** (G-827) of the range error times a Gaussian of the bearing error, because the two errors are taken as independent. Several landmarks multiply in the same way.

4.1 Two Gaussians, multiplied

The pink pole reads 3.10 m and 40°, but from pose A it should read 3.00 m and 36.87°: the real reading differs from the prediction by sensor noise. As for the measurement noise of a laser, small errors follow a **Gaussian** (G-827) centred on 0. Our sensor has these **standard deviations** (G-1871), the typical size of each error:

- σ_r is the standard deviation of the range error, 0.1 m;
- σ_φ is the standard deviation of the bearing error, 5°, which is 0.0873 rad.

We treat the range error and the bearing error as independent: knowing that the range came out 10 cm long says nothing about the bearing (Brown CS148 tutorial, Eq. 13). For independent events (G-934) the probabilities multiply, and so do the densities. So the density of one reading is the product of two Gaussians. For one reading and one guessed pose:

- r is the measured range, in m (3.10 m for the pink pole);
- $\hat{r}$ is the expected range from the guessed pose, in m (3.00 m from A);
- φ is the measured bearing (40° for the pink pole);
- $\hat{\varphi}$ is the expected bearing (36.87° from A);
- $\mathcal{N}(e;\ 0, \sigma^2)$ is the Gaussian density of an error e, with mean 0 and standard deviation σ;
- $p(z \mid x, m)$ is the density of the reading z given the guessed pose x and the map m, in per m per rad.

$$p(z \mid x, m) = \mathcal{N}(r - \hat{r};\ 0, \sigma_r^2)$$

$$\times\ \mathcal{N}(\varphi - \hat{\varphi};\ 0, \sigma_\varphi^2)$$

with the bearing difference wrapped. A camera can confuse colours, so Thrun adds a third Gaussian for the signature; with **known correspondence** (G-2405) that factor is the same for every pose and can be left out (Thrun §6.6; Brown CS148 tutorial, Eq. 16–17). The product of the two Gaussians is the **landmark measurement model** (G-2401) with known correspondence (Thrun §6.6, Table 6.4; Freiburg sensor-model slides, "Probabilistic Model").

Worked example: the pink pole at pose A. The errors:

$$r - \hat{r} = 3.10 - 3.00 = 0.10 \text{ m}$$

$$\varphi - \hat{\varphi} = 40° - 36.87° = 3.13°$$

$$3.13^\circ = 0.05463 \text{ rad}$$

The range Gaussian. Its peak height:

$$\frac{1}{\sigma_r\sqrt{2\pi}} = \frac{1}{0.1 \times 2.5066}$$

$$= 3.9895 \text{ per m}$$

The error is one standard deviation, so:

$$3.9895 \times e^{-\frac{1}{2}\times 1^2}$$

$$= 3.9895 \times 0.6065 = 2.420 \text{ per m}$$

The bearing Gaussian. Its peak height:

$$\frac{1}{\sigma_\varphi\sqrt{2\pi}} = \frac{1}{0.08727 \times 2.5066}$$

$$= 4.5714 \text{ per rad}$$

The error, in standard deviations:

$$\frac{0.05463}{0.08727} = 0.626$$

$$4.5714 \times e^{-\frac{1}{2}\times 0.626^2}$$

$$= 4.5714 \times 0.822 = 3.758 \text{ per rad}$$

The product:

$$2.420 \times 3.758 = 9.09$$

The result is a density, per metre per radian, so it can be larger than 1; only areas under it are probabilities (what a density means, G-1569). It is easy to call such a number "the probability of the reading", and wrong.

4.2 Several landmarks

Readings of different landmarks come from separate detections, so their errors are taken as independent too, and their densities multiply (Brown CS148 tutorial, Eq. 11). The three readings at pose A, with each pole's expected range $\hat{r}$ and expected bearing $\hat{\varphi}$:

Pole	$\hat{r}$ (m)	$\hat{\varphi}$	Range error	Bearing error	Density
pink	3.00	36.87°	0.10 m	3.13°	9.09
green	2.00	143.13°	−0.05 m	−2.13°	14.70
blue	2.00	−53.13°	0.05 m	−1.87°	15.01

The density of all three readings at A is the product of the three:

$$9.09 \times 14.70 \times 15.01 = 2006$$

At the wrong pose B = (3, 2, 0), one metre to the right, the pink pole should appear at 2.28 m and 52°, the green at 2.86 m and 155°, the blue at 1.61 m and −83°. Every reading misses by several **standard deviations** (G-1871), and the product is about 10^{-43}. The readings say A, overwhelmingly.

> **Python:** the model for one reading (Brown CS148 tutorial, Table 2).
>
> ```
> def landmark_density(pose, m, r, phi):
>     x, y, th = pose
>     r_hat = np.hypot(m[0] - x, m[1] - y)
>     phi_hat = np.arctan2(m[1] - y, m[0] - x) - th
>     return gauss(r - r_hat, 0.1) * gauss(wrap(phi - phi_hat),
>      ↪  np.radians(5))
> landmark_density((2, 2, 0), (4.4, 3.8), 3.10, np.radians(40))    # 9.09
> ```
>
> `gauss` is the **Gaussian** (G-827) density and `wrap` wraps an angle to $(-\pi, \pi]$; both are in `RO-012-landmark-measurement-model.ipynb`.

These scores judge poses the robot already guessed. The next section asks the reverse: which positions fit the readings at all?

5. Where can the robot be?

> **Key point:** One range reading allows every position on a ring around the landmark. Two rings meet in at most two points, and a third ring, or the map, picks one.

5.1 One landmark: a ring

Section 4 scored two guessed poses, A and B. Now turn the question around: the readings of Figure 1 are fixed, and we ask which robot positions fit them, starting with one. Take the pink pole's range, 3.10 m, alone. Every position 3.10 m from the pole predicts that range exactly, whatever its direction from the pole. So the positions that fit form a ring of radius

3.10 m around the pole, blurred by the 0.1 m noise (Freiburg sensor-model slides, “Distances only”).

Figure 4 draws the range **Gaussian** (G-827) over every position of the room as a surface: a curved ridge of height 3.99 per m along the ring, falling to zero on either side. The true position A sits on the ridge, and so does every other point of it.

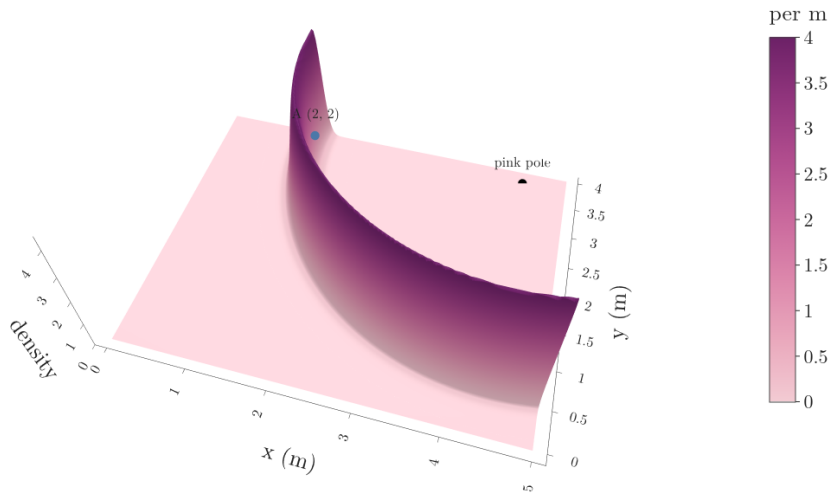


Figure 4: The density of the pink pole’s range reading over every robot position in the room, drawn as a surface: a curved ridge of radius 3.10 m around the pink pole, about 0.2 m wide, with the true position A on its crest

5.2 Two and three landmarks: rings meet

The green pole’s range, 1.95 m, gives a second ring. A position that fits both readings must lie on both rings. Two circles cross in at most two points, and here one of the two lies outside the room. Finding a position from distances to known points is **trilateration** (G-2406), the principle GPS uses with satellites.

Figure 5 multiplies the range densities in, one pole at a time; the product of densities is how readings combine, as in Section 4.2. Each frame is a surface like that of Figure 4 seen from above: darker colour means higher density, and each frame is divided by its own highest point, so its scale runs from 0 to 1. With pink alone the whole ring fits. With pink and green, only the spot near (1.9, 2.0) is left inside the room. With all three, the highest point is (1.90, 1.98), 10 cm from the true position, because each reading carries its own noise.

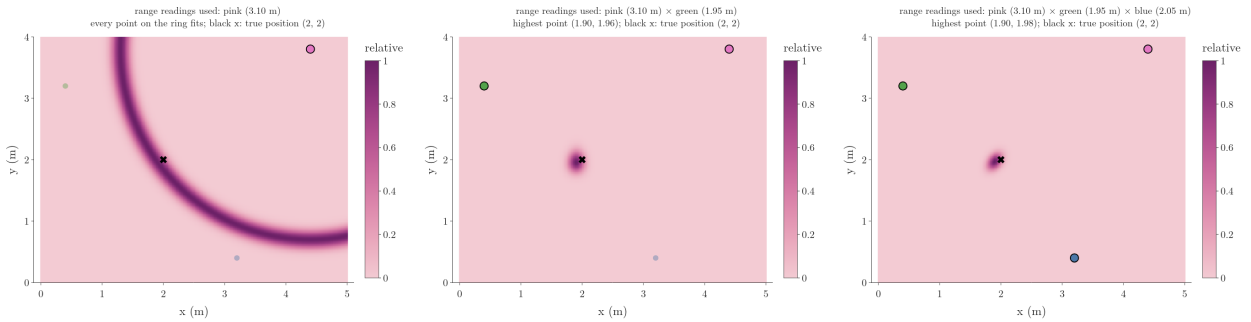


Figure 5: Animation over the room: first the density of the pink range reading, a ring around the pink pole; then pink times green, which leaves one bright spot near (1.9, 2.0); then pink times green times blue, one sharp spot at (1.90, 1.98) next to the true position (2, 2), marked by a black cross

The bearings add the **heading** (G-2284). Once the position is known, a bearing says which way the robot faces: the pole must appear at the measured angle from the heading. Rings and bearings say where the robot could be; Section 6 turns them into poses the robot can try.

6. Sampling poses from a landmark reading

> **Key point:** To propose poses that fit one reading, pick a random angle around the landmark, step out by the measured range, and turn the robot so the landmark appears at the measured bearing.

6.1 Why turn a reading around

The scores of Section 4 can only judge guesses the robot already has; they cannot help when no guess lies near A. A particle filter keeps a cloud of guessed poses and moves them with the motion model. If the robot is carried somewhere else, or starts with no idea where it is, none of its guesses may lie near the truth, and weighting them by the **measurement model** (G-2374) cannot fix that. A remedy is to add guesses drawn from the reading itself: poses where the robot could be, given what it just saw. Lenser and Veloso used exactly this on soccer-playing legged robots, adding poses drawn from the landmark readings whenever the filter was lost ("sensor resetting localization", Lenser and Veloso 2000). Drawing poses that fit a reading is **sampling poses from a measurement** (G-2408) (Thrun §6.6.3). The particle filter loop runs the filter itself; drawing random values from a distribution is sampling (G-2326).

6.2 The recipe, and where the heading comes from

Section 5.1 says the robot lies on a ring around the pole, at any angle. So we pick that angle at random, and add noise to the reading so the samples spread as the sensor's errors do (Thrun §6.6.3, Table 6.5). For one reading of the pink pole:

- r is the measured range, 3.10 m;

- φ is the measured bearing, 40°;
- σ_r is the standard deviation of the range noise, 0.1 m;
- σ_φ is the standard deviation of the bearing noise, 5°;
- $m_{c,x}$ is the pole's x coordinate in the map, 4.4 m;
- $m_{c,y}$ is its y coordinate in the map, 3.8 m;
- γ (gamma) is the direction from the pole to the robot, in degrees, drawn at random;
- $\tilde{r}$ is the range with noise added, in m;
- $\tilde{\varphi}$ is the bearing with noise added, in degrees;
- x is the sampled robot's x coordinate, in m;
- y is its y coordinate, in m;
- θ is the direction the sampled robot faces, in degrees.

The recipe:

1. Draw an angle γ (gamma) around the pole, uniformly from 0° to 360°.
2. Draw a noisy range and bearing:

$$\tilde{r} = r + \text{noise of std. dev. } \sigma_r$$

$$\tilde{\varphi} = \varphi + \text{noise of std. dev. } \sigma_\varphi$$

3. Place the robot at distance $\tilde{r}$ from the pole, in direction γ:

$$x = m_{c,x} + \tilde{r}\cos\gamma$$

$$y = m_{c,y} + \tilde{r}\sin\gamma$$

4. Set the **heading** (G-2284) (wrapped):

$$\theta = \gamma + 180^\circ - \tilde{\varphi}$$

Step 4 needs a reason. The robot sits in direction γ from the pole, so seen from the robot the pole lies the opposite way, half a turn further:

$$\text{direction to the pole} = \gamma + 180^\circ$$

The bearing is that direction minus the heading (Section 3.1):

$$\tilde{\varphi} = \gamma + 180^\circ - \theta$$

Solving for the heading gives step 4. Thrun writes the same step in radians, where π rad is 180°:

$$\theta = \gamma - \pi - \tilde{\varphi}$$

Both give the same heading, since adding 180° and subtracting 180° differ by a full turn.

Figure 6 runs the recipe on the noise-free reading of A, range 3.0 m and bearing 36.87°, one step at a time. Watch the angle γ swing round the pole to 216.87°, the robot step out 3.0 m to (2, 2), and its heading turn from the pole until the pole sits 36.87° to its left: the heading ends at 0°, so the recipe gives back pose A. The animation then runs the same four steps with the real reading and noise, 200 times (Section 6.3).

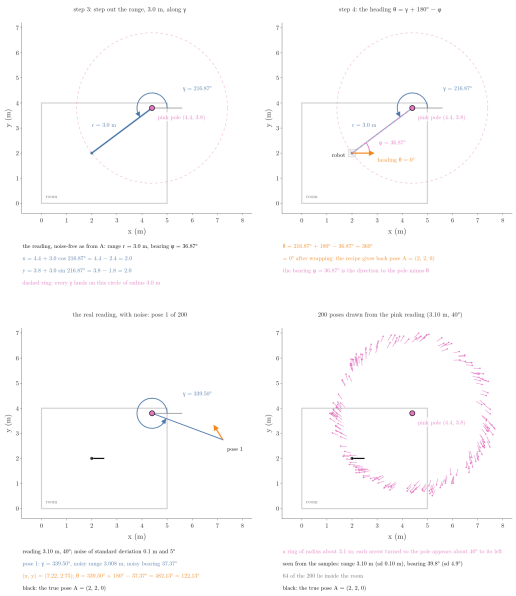


Figure 6: Animation in two parts. Part 1, one pose from the noise-free reading of A (range 3.0 m, bearing 36.87 degrees): the angle gamma around the pink pole swings from the grey 0-degree line to 216.87 degrees; the robot steps out 3.0 m along it to (2, 2), on the dashed circle of radius 3.0 m; seen from the robot the pole lies at 216.87 − 180 = 36.87 degrees, and the heading theta turns from there down to 0 degrees, so that the pole appears at the bearing phi = 36.87 degrees to its left; theta = 216.87 + 180 − 36.87 = 360, which is 0 after wrapping. Part 2, the real reading (3.10 m, 40 degrees) with noise of 0.1 m and 5 degrees: pose 1 with its numbers, then 5, 20, 50, 100 and 200 poses fill a ring of radius about 3.1 m around the pole, each arrow turned so that the pole appears about 40 degrees to its left; the black arrow is the true pose A. The last frame shows all 200 poses, 64 of them inside the room

Check with the noise-free reading. From the pink pole, A lies in the direction:

$$\gamma = \operatorname{atan2}(2 - 3.8,\ 2 - 4.4)$$

$$= -143.13°$$

Step 1 draws γ from 0° to 360°, so we add a full turn:

$$-143.13^\circ + 360^\circ = 216.87^\circ$$

With the predicted range 3.0 m and bearing 36.87°:

$$x = 4.4 + 3.0 \cos 216.87^\circ$$

$$= 4.4 - 2.4 = 2.0$$

$$y = 3.8 + 3.0 \sin 216.87^\circ$$

$$= 3.8 - 1.8 = 2.0$$

$$\theta = 216.87^\circ + 180^\circ - 36.87^\circ = 360^\circ$$

$$= 0^\circ \text{ after wrapping}$$

The recipe gives back pose A exactly. With the real reading (3.10 m, 40°) it gives (1.92, 1.94, −3.13°): close to A, off by the reading's own noise.

6.3 The samples

The second part of Figure 6 draws 200 poses from the pink reading, with noise. They form a ring of radius about 3.1 m around the pole, and each arrow is turned so that the pole appears about 40° to its left. Seen from the 200 samples, the pole's range averages 3.10 m with **standard deviation** (G-1871) 0.10 m and its bearing averages 39.8° with standard deviation 4.9°: the samples reproduce the reading and its noise, as they should.

Only 64 of the 200 samples lie inside the room. One reading does not know about walls; the map does. Samples outside the free space are thrown away, as the motion model does for poses inside walls, and readings of the other poles, through the **measurement model** (G-2374) of Section 4, then weight the rest.

7. Summary

The symbols in the table are listed in Section 3.1 (expected range and bearing), Section 4.1 (the density) and Section 6.2 (a pose from a reading).

Step	Formula	Why
expected range	$\hat{r} = \sqrt{\Delta x^2 + \Delta y^2}$	straight-line distance to the landmark's place in the map
expected bearing	$\hat{\varphi} = \operatorname{atan2}(\Delta y, \Delta x) - \theta$	the direction to the landmark, measured from the heading
density of a reading	$\mathcal{N}(r - \hat{r}) \times \mathcal{N}(\varphi - \hat{\varphi})$	range and bearing errors taken as independent
a pose from a reading	$(m_{c,x}, m_{c,y}) + \tilde{r}(\cos\gamma, \sin\gamma)$, $\theta = \gamma + 180° - \tilde{\varphi}$	the robot lies on a ring; the bearing fixes its heading

- The opening question has its answer: the three readings of Figure 1 score the true pose A at 2006 and the pose B one metre away at about 10^{-43}, because each pole, reduced to a range and a bearing, misses by several **standard deviations** (G-1871) from the wrong pose.
- A landmark reading is a range, a bearing and a signature; the signature gives the correspondence, so the robot knows which map landmark to predict.
- The expected bearing uses **atan2** (G-2325), because the plain arctangent of the ratio of the offsets, $\Delta y/\Delta x$, loses the signs and can point the opposite way ($-36.87°$ instead of $143.13°$ for the green pole).
- Bearing differences are wrapped into $-180°$ to $180°$, because $175°$ and $-178°$ are $7°$ apart, not $353°$.
- The density of a reading is the product of a range **Gaussian** (G-827) and a bearing Gaussian, because the two errors are taken as independent; readings of several landmarks multiply for the same reason. The product is a density, so it can exceed 1 (9.09 for the pink pole).
- One range reading fits a whole ring of positions, because every position at the measured distance predicts that range; two rings cross in at most two points, and a third reading or the map picks one; here the three poles put the robot within 10 cm of the truth.
- A reading can be turned around to draw poses that fit it: a random angle γ around the landmark, the measured range, and the **heading** (G-2284) that makes the landmark appear at the measured bearing, so that a lost particle filter can put guesses where the robot could really be.

The robot now has a motion model and sensor models for a laser and for landmarks, so it can score any guessed pose. Still missing: what it should keep in memory between steps, the **belief** (G-2433).

8. Sources

Built from

- Cyrill Stachniss, "Observation Models", YouTube, https://www.youtube.com/watch?v=SfwxLpdFB-o
- Carlotta A. Berry, PhD, "Advanced Mobile Robotics: Lecture 4-2a - Probabilistic Sensor Models", YouTube, https://www.youtube.com/watch?v=T8b2fMQbWug
- Schwertfeger, J. (2007). "Tutorial on a Probabilistic Measurement Model based on Landmark Range and Bearing Information". Brown University CS148. https://cs.brown.edu/courses/cs148/tutorials/measurement_model_tutorial.pdf (Brown CS148 tutorial)
- Burgard, W. et al., *Introduction to Mobile Robotics*, "Probabilistic Sensor Models" slides, University of Freiburg, http://ais.informatik.uni-freiburg.de/teaching/ss23/robotics/slides/07-sensor-models.pdf (Freiburg sensor-model slides)
- Lenser, S. and Veloso, M. (2000). "Sensor Resetting Localization for Poorly Modelled Mobile Robots". *IEEE International Conference on Robotics and Automation*. https://www.ri.cmu.edu/publications/sensor-resetting-localization-for-poorly-modelled-mobile-robots (Lenser and Veloso 2000)
- Thrun, S., Burgard, W. and Fox, D. (2005). *Probabilistic Robotics*. MIT Press. §6.6 (feature-based measurement models), §6.6.3 (sampling poses). Cited only for what the free sources above confirm or the Note derives. (Thrun)

Other references

- Oriolo, G., "Localization: Landmark-based and SLAM" slides, Sapienza University of Rome, https://www.diag.uniroma1.it/~oriolo/amr/slides/Localization3_Slides.pdf

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Known correspondence (G-2405)	Knowing which landmark of the map a reading belongs to, for example from a unique colour (its signature); with it the model can predict that one landmark's range and bearing.
Expected range and bearing $(\hat{r}, \hat{\varphi})$ (G-2402)	The range and bearing a landmark would show from a guessed pose: the straight-line distance to its place in the map, and its direction (by atan2) minus the robot's heading; the model compares the real reading with them.

Term	Meaning
Angle wrapping (G-2404)	Adding or subtracting full turns until an angle or angle difference lies between $-180°$ and $180°$, so that $175°$ and $-178°$ count as $7°$ apart, not $353°$; needed whenever bearings or headings are compared.
Landmark measurement model (G-2401)	The density of a landmark reading (range, bearing) given a pose and the map: a Gaussian of the range error times a Gaussian of the bearing error, with the errors taken as independent; it scores how well a guessed pose explains what the robot saw.
Trilateration (G-2406)	Finding a position from measured distances to points at known places: each distance puts the robot on a circle, and the circles meet at the position; GPS works this way.
Sampling poses from a measurement (G-2408)	Drawing poses that fit one reading: for a landmark, a random angle around it, the measured range plus noise, and the heading that makes the landmark appear at the measured bearing; used to add good guesses to a particle filter that is lost.
Gaussian distribution (G-827)	Another name for the normal distribution: the symmetric, bell-shaped continuous distribution set by its mean and standard deviation, used to model many measurements.
Likelihood field (G-2393)	A grid over the map that stores, for each cell, the density of a beam endpoint landing there; computed once from the distance transform, so scoring a pose becomes one look-up per beam.
Heading θ (G-2284)	The angle from the world x-axis to the direction a robot faces, measured counter-clockwise; it decides which way the robot moves when it drives forward.
Beam model (G-2377)	The measurement model of a range finder that mixes four densities for one reading (measurement noise, unexpected objects, failures, random readings) with weights that add up to 1, so no single odd reading can rule out the true pose.

Term	Meaning
Measurement model $p(z \mid x, m)$ (G-2374)	The density of getting sensor readings z if the robot were at pose x in map m (also sensor or observation model); read over poses it is a likelihood that tells a filter which poses fit the readings.
Range (of a landmark reading) (G-2343)	The distance from the robot to a sensed landmark, such as 2.236 m; together with the bearing it fixes where the landmark is relative to the robot.
Bearing (G-2344)	The angle from the robot's heading to a sensed landmark, positive to the left, such as 86.6 degrees; together with the range it fixes where the landmark is relative to the robot.
Signature (of a landmark) (G-2345)	A label that says what kind of landmark was sensed, such as its colour or ID number; it tells the robot which landmark in the map a reading belongs to.
Euclidean distance (G-715)	The straight-line distance between two points.
atan2 (G-2325)	A function atan2(rise, run) that returns the direction of a line over the full circle from its rise and run given separately; needed because the plain inverse tangent of the ratio cannot tell opposite directions apart.
Standard deviation (G-1871)	How far values typically lie from the mean; divide by $n - 1$ for a sample (pandas default) or by n for a population (NumPy default).
Independent events (G-934)	Events where one happening does not change the probability of the other.
Probability density (G-1569)	The height of a continuous distribution's curve; compares how likely nearby values are.
Sampling (from a distribution) (G-2326)	Drawing random values so that, over many draws, each value comes up as often as a given distribution says; it turns a motion model into a cloud of possible poses for particle filters.
Belief (G-2433)	The robot's probability for each possible state given all its commands and readings, $\mathrm{bel}(x_t) = p(x_t \mid z_{1:t}, u_{1:t})$; a short summary of the history that says how strongly the evidence points to each state.